



Practice Question

Assignment On

Quadratic Equation

Directions (Q.1 - 5): In the following questions, two equations I and II are given. You have to solve both the equations and give answer.

1) If $x > y$ 2) if $x \geq y$ 3) if $x < y$ 4) if $x \leq y$

5) if $x = y$ or the relationship cannot be determined

1. I) $x^2 - 12x + 35 = 0$ II. $y^2 - 9y + 20 = 0$

2. I) $4x + 3y = 16$ II. $2x + 2y = 9$

3. I) $x^2 + 7x + 12 = 0$ II. $y^2 + 5y + 6 = 0$

4. I) $x^2 + 9x + 20 = 0$ II) $y^2 + 5y + 6 = 0$

5. I) $2x + 4y = 13$ II) $4x + 2y = 14$

Directions (Q.6-10): In each question two equations are provided. On the basis of these you have to find out the relation between p and q. Give answer.

1) if $p = q$ 2) if $p > q$ 3) if $q > p$

4) if $p \geq q$ 5) if $q \geq p$

6. I. $p^2 + 7p + 10 = 0$ II. $q^2 - q - 6 = 0$

7. I. $2p^2 - 3p - 2 = 0$ II. $2q^2 - 11q + 15 = 0$

8. I. $2p^2 + 5p + 2 = 0$ II. $2q^2 + 11q + 14 = 0$

9. I. $p^2 - 19p + 88 = 0$ II. $q^2 - 48q + 576 = 0$

10. I. $p^2 + 2p - 8 = 0$ II. $q^2 - 5q + 6 = 0$

Directions (Q. 11-15): In each of these questions two equations numbered I and II are given. You have to solve both the equations and give answer, if –

(1) $= q$, (2) $p > q$, (3) $p < q$,

(4) $p \geq q$, (5) $p \leq q$

11. I. $18p^2 - 21p + 6 = 0$

II. $16q^2 - 24q + 9 = 0$

12. I. $\frac{9}{4}p + \frac{7}{8}p = 2\frac{1}{12}$

II. $\frac{7}{5}p = \frac{9}{10}q + \frac{1}{4}$

13. I. $\frac{8}{21}p - \frac{2}{7} = 0$

II. $16q^2 - 24q + 9 = 0$

14. I. $3q^2 - 29q + 18 = 0$

II. $9p^2 - 4 = 0$

15. I. $p^2 - 10p + 21 = 0$

II. $q^2 + q - 12 = 0$

Directions (Q.16-20): In the following questions, two equations numbered I and II are given. You have to solve both the equations and find out the correct option. Give answer

(1) If $x \geq y$ (2) If $x \leq y$ (3) If $x > y$ (4) If $x < y$

(5) If $x = y$ or relationship between x and y cannot be established

16. I. $6x^2 - 49x + 99 = 0$

II. $5y^2 + 17y + 14 = 0$

17. I. $5x^2 - 19x = -12$

II. $5y^2 + 11y - 12$

18. I. $x = (1331)^{\frac{1}{3}}$

II. $2y^2 - 21y + 55 = 0$

19. I. $5x = 7y + 21$

II. $11x + 4y + 109 = 0$

20. I. $2x^2 - 11x + 12 = 0$

II. $2y^2 - 17y + 36 = 0$

Directions (Q.21-25): In the following questions, two equations numbered I and II are given. You have to solve both the equations and find out the correct option. Give answer

(1) If $x \geq y$ (2) If $x \leq y$ (3) If $x > y$ (4) If $x < y$

(5) If $x = y$ or relationship between x and y cannot be established

21. I. $2x^2 - 15x + 13 = 0$ II. $3y^2 - 6y + 3 = 0$

22. I. $2x^2 + x - 1 = 0$ II. $2y^2 - 3y + 1 = 0$

23. I. $4x^2 - 4x + 1 = 0$ II. $4y^2 + 4y + 1 = 0$

24. I. $9x^2 + 6x + 1 = 0$ II. $9y^2 - 6y + 1 = 0$

25. I. $x^2 + 5x + 6 = 0$ II. $y^2 + 7y + 12 = 0$

Directions: (Q26-30) In the following questions two equations numbered I and II are given. You have to solve both the equations. Give answer if;

1) if $x = y$ or relationship between x and y cannot be established

2) if $x > y$ 3) if $x < y$ 4) if $x \geq y$ 5) if $x \leq y$

26. I. $x^2 = 10$ II. $y^2 - 9y + 20 = 0$

27. I. $64x^2 - 64x + 15 = 0$ II. $21y^2 - 13y + 2 = 0$

28. I. $x^2 - 2x - \sqrt{5}x + 2\sqrt{5} = 0$ II. $y^2 - \sqrt{3}y - \sqrt{2}y + \sqrt{6} = 0$

29. $2x^2 + 5x + 2 = 0$ II. $4y^2 = 1$

30. I. $3x^2 - 14x + 15 = 0$ II. $15y^2 - 34y + 15 = 0$

Directions (Q.31-35): In the following questions, two equations numbered I and II are given. You have to solve both the equations and give answer.

(i) if $x > y$ (ii) if $x \geq y$ (iii) $x < y$ (iv) $x \leq y$

(v) if $x = y$ or the relationship cannot be established

31. I. $3y^2 - 7y + 2 = 0$ II. $24x^2 + 26x - 5 = 0$

32. I. $8y^2 - 6y - 9 = 0$ II. $6x^2 - x - 12 = 0$

33. I. $10x^2 - 11x + 3 = 0$ II. $42y^2 - 23y - 10 = 0$

34. I. $4x^2 - 15x + 9 = 0$ II. $20y^2 - 23y + 6 = 0$

35. I. $18x^2 + 27x + 10 = 0$ II. $12y^2 + 17y + 6 = 0$

Directions (Q.36-40): In the following questions, two equations numbered I and II are given. You have to solve both the equations and give answer.

(i) if $x > y$ (ii) if $x \geq y$ (iii) $x < y$ (iv) $x \leq y$

(v) if $x = y$ or the relationship cannot be established

36. I. $8x^2 + 2x - 3 = 0$ II. $6y^2 + 11y + 4 = 0$

37. I. $2x^2 - 9x + 10 = 0$ II. $3y^2 + 11y + 6 = 0$

38. I. $6x^2 - 23x + 7 = 0$ II. $6y^2 - 29y + 9 = 0$

39. I. $20y^2 - 13y + 2 = 0$ II. $6x^2 - 25x + 14 = 0$

40. I. $2x^2 - 9x + 9 = 0$ II. $2y^2 - 7y + 3 = 0$

Directions (Q41- 45) In the following questions two equations numbered I and II are given. You have to solve both the equations. Give answer if;

(1) $x < y$, **(2)** $x > y$,

(3) $x = y$ or no relationship cannot be established between x and y .

(4) $x \geq y$, **(5)** $x \leq y$

41. I. $x^2 = 10$

II. $y^2 - 9y + 20 = 0$

42. I. $2x^2 - 15x + 27 = 0$

II. $2y^2 - 13y + 20 = 0$

43. I. $9x^2 - 21x + 10 = 0$

II. $y^2 - 8y + 15 = 0$

44. I. $2x^2 - 13x + 15 = 0$

II. $2y^2 - 11y + 12 = 0$

45. I. $2x^2 + 7x + 6 = 0$

II. $2y^2 + 17y + 30 = 0$

Directions (Q.46-50): In the following questions, two equations numbered I and II are given. You have to solve both the equations and give answer.

(i) if $x > y$

(ii) if $x \geq y$

(iii) $x < y$

(iv) $x \leq y$

(v) if $x = y$ or the relationship cannot be established

46. I. $8x^2 - 14x + 3 = 0$

II. $8y^2 - 22y + 5 = 0$

47. I. $2x^2 - 11x + 12 = 0$

II. $12y^2 + 29y + 15 = 0$

48. I. $2y^2 - 3y - 14 = 0$

II. $3x^2 - 7x + 4 = 0$

49. I. $6x^2 + 37x + 45 = 0$

II. $3y^2 - 11y + 6 = 0$

50. I. $7x^2 + 52x + 21 = 0$

II. $6y^2 + 7y - 24 = 0$

Directions (Q.51-55): In the following questions, two equations numbered I and II are given. You have to solve both the equations and give answer.

(i) if $x > y$ (ii) if $x \geq y$ (iii) $x < y$ (iv) $x \leq y$

(v) if $x = y$ or the relationship cannot be established

51. I. $35x^2 - 24x - 35 = 0$

II. $72y^2 - 145y + 72 = 0$

52. I. $77x^2 - 25x - 72 = 0$

II. $42y^2 + 13y - 42 = 0$

53. I. $35x^2 - 51x + 18 = 0$

II. $30y^2 + 17y - 21 = 0$

54. I. $117x^2 + 250x + 117 = 0$

II. $54y^2 - 123y + 65 = 0$

55. I. $56x^2 - 99x + 40 = 0$

II. $8y^2 - 30y + 25 = 0$

Directions (Q56-60): In the following questions, two equations numbered I and II are given. You have to solve both the equations and give answer.

(i) if $x > y$ (ii) if $x \geq y$ (iii) $x < y$ (iv) $x \leq y$

(v) if $x = y$ or the relationship cannot be established

56. I. $4x^2 - 21x + 20 = 0$

II. $8y^2 - 22y + 15 = 0$

57. I. $y^2 - 7y - 18 = 0$

II. $x^2 + 10x + 16 = 0$

58. I. $2y^2 - 11y + 15 = 0$

II. $2x^2 + 3x - 14 = 0$

59. I. $3x^2 - 5x + 2 = 0$

II. $6y^2 - 19y + 14 = 0$

60. I. $6y^2 + 11y - 7 = 0$

II. $21x^2 + 5x - 6 = 0$

Directions : (Q.61- 65) In the following questions two equations numbered I and II are given. You have to solve both the equations. Give answer if;

- (1) $x < y$, (2) $x > y$,
 (3) $x = y$ or no relationship cannot be established between x and y
 (4) $x \geq y$, (5) $x \leq y$

61. I. $17x^2 - 26x - 16 = 0$

II. $17y^2 - 26y + 9 = 0$

62. I. $24x^2 - 58x + 23 = 0$

II. $20y^2 + 24y - 65 = 0$

63. I. $x^{\frac{7}{3}} \div 27 = 512 \div x^{\frac{2}{3}}$

II. $y^{\frac{1}{5}} \times y^{\frac{1}{5}} \times 16 = 1296 \div y^{\frac{3}{5}}$

64. I. $3x^6 - 19x^3 + 16 = 0$

II. $9y^4 - 27y^2 + 20 = 0$

65. I. $\sqrt{2x+9} + x = 13$

II. $\sqrt{4y^2 - 4y + 1} - 2y + 6 = 0$

Directions (Q66-70) in each of these questions, two equations numbered I and II are given. You have to solve both the equation and mark the appropriate option – give answer

- 1) If $x > y$ 2) If $x \geq y$ 3) If $x < y$ 4) If $x \leq y$
 5) If $x = y$ or relationship between x and y cannot be established.

66. I. $x = \sqrt{16102 - 3333}$

II. $y^2 = 12769$

67. I. $4x^2 + 3\sqrt{7}x - 7 = 0$ II. $7y^2 + 4\sqrt{7}y - 5 = 0$

68. I. $7x + 11y - 141 = 0$ II. $9y - 11x - 15 = 0$

69. I. $6x^2 - 41x + 13 = 0$ II. $2y^2 - 19y + 42 = 0$

70. I. $x = \sqrt{20 + \sqrt{20 + \sqrt{20 + \sqrt{20} \dots \dots \dots \infty}}}$

II. $y = \sqrt{5\sqrt{5\sqrt{5\sqrt{5} \dots \dots \dots \infty}}}$

Directions (Q.71-75): In the following questions, two equations numbered I and II are given. You have to solve both the equations and find out the correct option. Give answer

1) If $x \geq y$ 2) If $x \leq y$ 3) If $x > y$ 4) If $x < y$

5) If $x = y$ or relationship between x and y cannot be established

71. I. $\sqrt{289}x + \sqrt{25} = 0$

II. $\sqrt{676}y + 10 = 0$

72. I. $8x^2 - 78x + 169 = 0$

II. $20y^2 - 117y + 169 = 0$

73. I. $\frac{15}{\sqrt{x}} + \frac{9}{\sqrt{x}} = 11\sqrt{x}$

II. $\frac{\sqrt{y}}{4} + \frac{5\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

74. I. $\frac{8}{\sqrt{x}} + \frac{6}{\sqrt{x}} = \sqrt{x}$

II. $y^3 - \frac{(14)^{\frac{7}{2}}}{\sqrt{y}} = 0$

75. I. $x^2 - 208 = 233$

II. $y^2 + 47 - 371 = 0$

Directions (Q76- 80) In the following questions two equations numbered I and II are given. You have to solve both the equations. Give answer if;

(1) $x < y$,

(2) $x > y$,

(3) $x = y$ or no relationship cannot be established between x and y .

(4) $x \geq y$,

(5) $x \leq y$

76. I. $90x^2 - 113x + 35 = 0$

II. $88y^2 - 149y + 63 = 0$

77. I. $35x^2 + 83x + 36 = 0$

II. $42y^2 + 53y + 15 = 0$

78. I. $96x^2 + 52x - 63 = 0$

II. $77y^2 + 155y + 72 = 0$

79. I. $20x^2 - 93x + 108 = 0$

II. $72y^2 - 47y - 144 = 0$

80. I. $9x^2 + 6\sqrt{7}x + 7 = 0$

II. $4y^2 + 4\sqrt{3}y + 3 = 0$

Directions (Q81-85): In the following questions two equations numbered I and II are given. You have to solve both the equations. Give answer :

(1) If $x < y$, (2) If $x > y$,

(3) If $x = y$ or no relationship cannot be established between x and y

(4) If $x \geq y$, (5) If $x \leq y$

81. I. $143y^2 - 161y + 36 = 0$ II. $36x^2 - 161x + 143 = 0$

82. I. $84y^2 - 187y + 104 = 0$ II. $132x^2 - 41x - 40 = 0$

83. I. $63x^2 + 146x + 80 = 0$ II. $42y^2 + 109y + 70 = 0$

84. I. $70x^2 - 227x + 117 = 0$ II. $72y^2 + 23y - 35 = 0$

85. I. $40x^2 + 81x + 35 = 0$ II. $63y^2 + 103y + 42 = 0$

Directions (Q86- 90): In the following questions two equations numbered I and II are given. You have to solve both the equations. Give answer :

(1) If $x < y$,

(2) If $x > y$,

(3) If $x = y$ or no relationship cannot be established between x and y

(4) If $x \geq y$,

(5) If $x \leq y$

86. I. $35x^2 - 46x - 16 = 0$ II. $35y^2 - 116y + 96 = 0$

87. I. $195x^2 - 46x - 21 = 0$ II. $209y^2 + 7y - 12 = 0$

88. I. $165x^2 + 97x + 10 = 0$ II. $117y^2 - 163y + 56 = 0$

89. I. $144x^2 - 163x - 65 = 0$ II. $91y^2 - 128y - 48 = 0$

90. I. $84x^2 - 167x - 55 = 0$ II. $247y^2 + 210y + 27 = 0$

Directions (Q91- 95): In the following questions two equations numbered I and II are given. You have to solve both the equations. Give answer :

(1) fI $x < y$, (2) fI $x > y$,

(3)fI $x = y$ or no relationship cannot be established between x and y

(4)fI $x \geq y$, (5)fI $x \leq y$

91. I. $27x^2 + 120x + 77 = 0$

II. $56y^2 + 117y + 36 = 0$

92. I. $104x^2 + 9x - 35 = 0$

II. $72y^2 - 85y + 25 = 0$

93. I. $99x^2 + 161x + 26 = 0$

II. $26y^2 + 161y + 99 = 0$

94. $70x^2 - 143x + 72 = 0$

II. $80y^2 - 142y + 63 = 0$

95. I. $99x^2 + 31x - 110 = 0$

II. $6y^2 - 31y + 35 = 0$

Directions (Q.96-100): In the following questions, two equations numbered I and II are given.

You have to solve both the equations and find out the correct option. Give answer

(1) If $x \geq y$ (2) If $x \leq y$ (3) If $x > y$ (4) If $x < y$

(5) If $x = y$ or relationship between x and y cannot be established

96. I. $2x^2 - 12x + 16 = 0$

II. $4y^2 - 8y - 12 = 0$

97. I. $5x^2 - 24x + 28 = 0$

II. $4y^2 - 8y - 12 = 0$

98. I. $5x^2 - 14x + 8 = 0$

II. $2y^2 + 17y + 36 = 0$

99. I. $3x^2 - 22x + 40 = 0$

II. $4y^2 + 22y + 24 = 0$

100. I. $2x^2 + 11x + 15 = 0$

II. $2y^2 - 19y + 44 = 0$

Solution

Sol- 1. 2;

I. $x^2 - 12x + 35 = 0$, or $x^2 - 7x - 5x + 35 = 0$

Or, $x(x - 7) - 5(x - 7) = 0$, Or, $x = 5$ or 7

II. $y^2 - 9y + 20 = 0$, or $y^2 - 5y - 4y + 20 = 0$

Or $y(y - 5) - 4(y - 5) = 0$, $y = 4$ or 5 ,

Hence, $x \geq y$.

Sol-2. 1;

I. $4x + 3y = 16$ (1)

II. $2x + 2y = 9$; $\Rightarrow 4x + 4y = 18$ (2),

From (1) and (2), we get, $-y = -2$ or, $y = 2$,

Now, using 1, we get

$4x + 3 \times 2 = 16$, or $4x = 10$ or, $x = 2.5$

Hence, $x > y$.

Sol-3. 4; I. $x^2 + 7x + 12 = 0$, or $x^2 + 4x + 3x + 12 = 0$

Or, $x(x + 4) + 3(x + 4) = 0$, or Or $x = -3$ or -4

II. $y^2 + 5y + 6 = 0$ or, $y^2 + 3y + 2y + 6 = 0$

Or, $y(y + 3) + 2(y + 3) = 0$ or, $y = -2$ or -3 ,

Hence, $x \leq y$.

Assignment On Quadratic Equation

Sol-4. 3;

$$I. \Rightarrow x^2 + 9x + 20 = (x+4)(x+5)=0$$

$$\Rightarrow x = -4, -5;$$

$$II \Rightarrow y^2 + 5y + 6 = (y+2)(y+3)=0$$

$$\Rightarrow y = -2, -3,$$

Hence, $x < y$.

Sol-5. 1;

$$I + II \Rightarrow x=2.5 \text{ and } y = 2,$$

Hence, $x > y$.

Sol-6. 5;

$$I. p^2 + 5p + 2p + 10 + 6 = 0; \text{ or } p(p+5) + 2(p+5) = 0$$

$$(p+2)(p+5) = 0, \text{ or } p = -2 \text{ or } -5,$$

$$II. q^2 - q - 6 = 0; q^2 - 3q + 2q - 6 = 0$$

$$q(q-3) + 2(q-3) = 0 \text{ or, } q = -2 \text{ or } 3$$

Hence, $q \geq p$.

Assignment On Quadratic Equation

Sol-7. 3;

$$I. 2p^2 - 3p - 2 = 0, \text{ or, } (2p + 1)(p - 2) = 0$$

$$\text{Or, } p = -\frac{1}{2} \text{ or } 2,$$

$$II. 2q^2 - 11q + 15 = 0$$

$$\text{Or, } (2q - 5)(q - 3) = 0, q = \frac{5}{2} \text{ or } 3;$$

Hence, $q > p$.

Sol-8. 4;

$$I. 2p^2 + 5p + 2 = 0; (2p + 1)(p + 2) = 0$$

$$\text{or, } p = -\frac{1}{2} \text{ or } -2,$$

$$II. 2q^2 + 11q + 14 = 0$$

$$(q + 2)(2q + 7) = 0 \text{ or } q = -2 \text{ or } -\frac{7}{2}$$

Hence, $p \geq q$.

Sol-9. 3;

$$I. p^2 - 19p + 88 = 0$$

$$(p - 11)(p - 8) = 0 \Rightarrow p = 11 \text{ or } 8$$

$$II. q^2 - 48q + 576 = 0$$

$$(q - 24)(q - 24) = 0 \Rightarrow q = 24 \text{ or } 24;$$

Hence, $q > p$

Assignment On Quadratic Equation

Sol-10. 5;

$$I. p^2 + 2p - 8 = 0;$$

$$\Rightarrow p^2 + 4p - 2p - 8 = 0$$

$$\Rightarrow (p + 4)(p - 2) = 0$$

$$\Rightarrow p = 2, -4$$

$$II. q^2 - 5q + 6 = 0$$

$$\Rightarrow (q - 3)(q - 2) = 0$$

$$\Rightarrow q = 2, -3$$

Hence, $q \geq p$

Sol-11. 3;

$$I. 18p^2 - 21p + 6 = 0$$

$$\Rightarrow 6p^2 - 7p + 2 = 0$$

$$\Rightarrow 6p^2 - 4p - 3p + 2 = 0$$

$$\Rightarrow 2p(3p - 2) - 1(3p - 2) = 0$$

$$\Rightarrow (2p - 1)(3p - 2) = 0$$

$$\Rightarrow p = \frac{1}{2}, \frac{2}{3}$$

$$II. 16q^2 - 24q + 9 = 0$$

$$\Rightarrow (4q - 3)^2 = 0$$

$$\Rightarrow q = \frac{3}{4}$$

$$\therefore p < q$$

Sol-12. 3

$$I. \frac{9}{4}p + \frac{7}{8}p = 2\frac{1}{12}$$

$$\Rightarrow \frac{25}{8}p = \frac{25}{12}$$

$$\Rightarrow p = \frac{2}{3}$$

$$II. \frac{7}{5}p = \frac{9}{10}q + \frac{1}{4}$$

$$\Rightarrow \frac{7}{5} \times \frac{2}{3} - \frac{1}{4} = \frac{9}{10}q$$

$$\Rightarrow q = \frac{41}{54}$$

$$\therefore p < q$$

Sol-13. 1

$$I. \frac{8}{21}p - \frac{2}{7} = 0 \Rightarrow p = \frac{2}{7} \times \frac{21}{8} = \frac{3}{4}$$

$$II. 16q^2 - 24q + 9 = 0$$

$$\Rightarrow (4q - 3)^2 = 0$$

$$\Rightarrow q = \frac{3}{4}$$

$$\therefore p = q$$

Assignment On Quadratic Equation

Sol-14. 5

$$I. 3q^2 - 29q + 18 = 0$$

$$3q^2 - 2q - 27q + 18 = 0$$

$$\Rightarrow q(3q - 2) - 9(3q - 2) = 0$$

$$\Rightarrow (3q - 2)(q - 9) = 0$$

$$\Rightarrow q = 9, \frac{2}{3}$$

$$II. 9p^2 - 4 = 0$$

$$\Rightarrow p^2 = \frac{4}{9}$$

$$\Rightarrow p = \pm \frac{2}{3}$$

$$\therefore p \leq q$$

Sol-15. 4

$$I. p^2 - 10p + 21 = 0$$

$$p^2 - 3p - 7p + 21 = 0$$

$$\Rightarrow p(p - 3) - 7(p - 3) = 0$$

$$\Rightarrow (p - 7)(p - 3) = 0$$

$$\Rightarrow p = 3, 7$$

$$II. q^2 + q - 12 = 0$$

$$\Rightarrow q^2 + 4q - 3q - 12 = 0$$

$$\Rightarrow q(q + 4) - 3(q + 4) = 0$$

$$\Rightarrow q = -4, 3$$

$$\therefore p \geq q$$

Sol-16. 3;

I. $6x^2 - 49x + 99 = 0$

$\Rightarrow (3x - 11)(2x - 9) = 0$

$\Rightarrow x = \frac{11}{3}, \frac{9}{2}$

II. $5y^2 + 17y + 14 = 0$

$\Rightarrow (5y + 7)(y + 2) = 0$

$\Rightarrow y = -\frac{7}{5}, -2$

Hence, $x > y$

Sol-17. 1

I. $5x^2 - 19x = -12$

$\Rightarrow x = 3, \frac{4}{5}$

II. $5y^2 + 11y - 12$

$\Rightarrow y = -3, \frac{4}{5}$

Hence, $x \geq y$

Sol-18. 3;

I. $x = (1331)^{\frac{1}{3}}$

$\Rightarrow x = 11$

II. $2y^2 - 21y + 55 = 0$

$(2y - 11)(y - 5) = 0$

$\Rightarrow y = 5, \frac{11}{2}$

Hence, $x > y$

Assignment On Quadratic Equation

Sol-19. 3;

I. $5x = 7y + 21$

II. $11x + 4y + 109 = 0$

Solving these equations

$\Rightarrow x = -7$ and $y = -8$

Hence, $x > y$

Sol-20. 2;

I. $2x^2 - 11x + 12 = 0$

$\Rightarrow x = 4, \frac{3}{2}$

II. $2y^2 - 17y - 36 = 0$

$\Rightarrow y = 4, \frac{9}{2}$

Hence, $x \leq y$

Sol-21. 1

I. $2x^2 - 15x + 13 = 0$

$2x^2 - 2x - 13x + 13 = 0$

$2x(x - 1) - 13(x - 1)$

$(2x - 13)(x - 1) = 0$

$\Rightarrow x = 1, \frac{13}{2}$

II. $3y^2 - 6y + 3 = 0$

$3y^2 - 3y - 3y + 3 = 0$

$3y(y - 1) - 3(y - 1)$

$(3y - 3)(y - 1) = 0$

$\Rightarrow y = 1, 1$

Hence, $x \geq y$

Assignment On Quadratic Equation

Sol-22. 2;

$$I. 2x^2 + x - 1 = 0$$

$$(x - 1)(2x + 1) = 0$$

$$\Rightarrow x = 1, -\frac{1}{2}$$

$$II. 2y^2 - 3y + 1 = 0$$

$$(2y - 1)(y - 1) = 0$$

$$\Rightarrow y = 1, \frac{1}{2}$$

Hence, $x \leq y$

Sol-23. 3;

$$I. 4x^2 - 4x + 1 = 0$$

$$(2x - 1)^2 = 0$$

$$\Rightarrow x = \frac{1}{2}$$

$$II. 4y^2 + 4y + 1 = 0$$

$$(2y + 1)^2 = 0$$

$$\Rightarrow y = -\frac{1}{2}$$

Hence, $x > y$

Assignment On Quadratic Equation

Sol-24. 4;

$$I. 9x^2 + 6x + 1 = 0$$

$$(3x + 1)^2 = 0$$

$$\Rightarrow x = -\frac{1}{3}$$

$$II. 9y^2 - 6y + 1 = 0$$

$$(3y - 1)^2 = 0$$

$$\Rightarrow y = \frac{1}{3}$$

Hence, $x < y$

Sol-25. 1;

$$I. x^2 + 5x + 6 = 0$$

$$(x + 3)(x + 2) = 0$$

$$\Rightarrow x = -3, -2$$

$$II. y^2 + 7y + 12 = 0$$

$$(y + 4)(y + 3) = 0$$

$$\Rightarrow y = -3, -4$$

Hence, $x \geq y$

Sol-26. 3

I. $x^2 = 10$

I. $x = \pm \sqrt{10}$

II. $y^2 - 4y + 20 = 0$

II. $y^2 - 5y - 4y + 20 = 0$

II. $y(y - 5) - 4(y - 5) = 0$

$(y - 4)(y - 5) = 0$

$y = 4, 5$

Hence, $x < y$

Sol-27. 2

I. $64x^2 - 64x + 15 = 0$

$64x^2 - 24x - 40x + 15 = 0$

$8x(8x - 3) - 5(8x - 3) = 0$

$(8x - 3)(8x - 5) = 0$

$x = \frac{3}{8}, \frac{5}{8}$

II. $21y^2 - 13y + 2 = 0$

$21y^2 - 7y - 6y + 2 = 0$

$7y(3y - 1) - 2(3y - 1) = 0$

$y = \frac{2}{7}, \frac{1}{3}$

Hence, $x > y$

Sol-28. 2

I. $x^2 - 2x - \sqrt{5}x + 2\sqrt{5} = 0$

$$x(x - 2) - \sqrt{5}(x - 2) = 0$$

$$(x - 2)(x - \sqrt{5}) = 0$$

$$x = 2, \sqrt{5}$$

II. $y^2 - \sqrt{3}y - \sqrt{2}y + \sqrt{6} = 0$

$$y(y - \sqrt{3}) - \sqrt{2}(y - \sqrt{3}) = 0$$

$$(y - \sqrt{3})(y - \sqrt{2}) = 0$$

$$y = \sqrt{3}, \sqrt{2}$$

Hence, $x > y$

Sol-29. 5

I. $2x^2 + 2x + 3x + 2 = 0$

$$2x^2 + 4x + x + 2 = 0$$

$$2x(x + 2) + 1(x + 2) = 0$$

$$(2x + 1)(x + 2) = 0$$

$$x = -\frac{1}{2}, -2$$

II. $4y^2 = 1$

$$y^2 = \frac{1}{4}$$

$$y = \pm \frac{1}{2}$$

Hence, $x \leq y$

Sol-30. 4

I. $3x^2 - 14x + 15 = 0$

$$3x^2 - 9x - 5x + 15 = 0$$

$$3x(x - 3) - 5(x - 3) = 0$$

$$(x - 3)(3x - 5) = 0$$

$$x = 3, \frac{5}{3}$$

II. $15y^2 - 34y + 15 = 0$

$$5y^2 - 25y - 9y + 15 = 0$$

$$5y(3y - 5) - 3(3y - 5) = 0$$

$$(3y - 5)(5y - 3) = 0$$

$$y = \frac{5}{3}, \frac{3}{5}$$

Hence, $x \geq y$

Sol-31. 3

I. $3y^2 - 7y + 2 = 0$

$$3y^2 - y - 6y + 2 = 0$$

$$y(3y - 1) - 2(3y - 1) = 0$$

$$(y - 2)(3y - 1) = 0$$

$$\therefore y = 2, \frac{1}{3}$$

II. $24x^2 + 26x - 5 = 0$

$$24x^2 - 4x + 30x - 5 = 0$$

$$4x(6x - 1) + 5(6x - 1) = 0$$

$$(4x + 5)(6x - 1) = 0$$

$$\therefore x = \frac{1}{6}, -\frac{5}{4}$$

Hence, $x < y$

Sol-32. 5

I. $8y^2 - 6y - 9 = 0$

$8y^2 + 12y - 6y - 9 = 0$

$4y(2y + 3) - 3(2y + 3) = 0$

$(4y - 3)(2y + 3) = 0$

$\therefore y = -\frac{3}{2}, \frac{3}{4}$

II. $6x^2 - x - 12 = 0$

$6x^2 - 9x + 8x - 12 = 0$

$3x(2x - 3) + 4(2x - 3) = 0$

$(2x - 3)(3x + 4) = 0$

$(2x - 3)(3x + 4) = 0$

$\therefore x = \frac{3}{2}, -\frac{4}{3}$

Hence, relationship cannot be established between x and y

Sol-33. 5

I. $10x^2 - 11x + 3 = 0$

$10x^2 - 6x - 5x + 3 = 0$

$2x(5x - 3) - 1(5x - 3) = 0$

$(2x - 1)(5x - 3) = 0$

$\therefore x = \frac{1}{2}, \frac{3}{5}$

II. $42y^2 - 23y - 10 = 0$

$42y^2 - 35y + 12y - 10 = 0$

$7y(6y - 5) + 2(6y - 5) = 0$

$(7y + 2)(6y - 5) = 0$

$\therefore y = -\frac{2}{7}, \frac{5}{6}$

Hence, relationship cannot be established between x and y

Sol-34. 2

I. $4x^2 - 15x + 9 = 0$

$4x^2 - 12x - 3x + 9 = 0$

$4x(x - 3) - 3(x - 3) = 0$

$(4x - 3)(x - 3) = 0$

$\therefore x = 3, \frac{3}{4}$

II. $20y^2 - 23y + 6 = 0$

$20y^2 - 8y - 15y + 6 = 0$

$4y(5y - 2) - 3(5y - 2) = 0$

$(4y - 3)(5y - 2) = 0$

$\therefore y = \frac{2}{5}, \frac{3}{4}$

Hence, $x \geq y$

Sol-35. 4

I. $18x^2 + 27x + 10 = 0$

$18x^2 + 12x + 15x + 10 = 0$

$6x(3x + 2) + 5(3x + 2) = 0$

$(6x + 5)(3x + 2) = 0$

$\therefore x = -\frac{5}{6}, -\frac{2}{3}$

II. $12y^2 + 17y + 6 = 0$

$12y^2 + 9y + 8y + 6 = 0$

$3y(4y + 3) + 2(4y + 3) = 0$

$(3y + 2)(4y + 3) = 0$

$\therefore y = -\frac{2}{3}, -\frac{3}{4}$

Hence, relationship cannot be established between x and y

Assignment On Quadratic Equation

Sol-36. 5

I. $8x^2 + 2x - 3 = 0$

$$8x^2 + 6x - 4x - 3 = 0$$

$$2x(4x + 3) - 1(4x + 3) = 0$$

$$(2x - 1)(4x + 3) = 0$$

$$x = \frac{1}{2}, -\frac{3}{4}$$

II. $6y^2 + 11y + 4 = 0$

$$6y^2 + 8y + 3y + 4 = 0$$

$$2y(3y + 4) + 1(3y + 4) = 0$$

$$(2y + 1)(3y + 4) = 0$$

$$y = -\frac{1}{2}, -\frac{4}{3}$$

Hence, no relationship cannot be established between x and y

Sol-37. 1

I. $2x^2 - 9x + 10 = 0$

$$2x^2 - 5x - 4x + 10 = 0$$

$$x(2x - 5) - 2(2x - 5) = 0$$

$$(x - 2)(2x - 5) = 0$$

$$x = 2, \frac{5}{2}$$

II. $3y^2 + 11y + 6 = 0$

$$3y^2 + 2y + 9y + 6 = 0$$

$$y(3y + 2) + 3(3y + 2) = 0$$

$$y = -3, -\frac{2}{3}$$

Hence, $x > y$

Sol-38. 4

I. $6x^2 - 23x + 7 = 0$

$6x^2 - 2x - 21x + 7 = 0$

$2x(3x - 1) - 7(3x - 1) = 0$

$(2x - 7)(3x - 1) = 0$

$x = \frac{7}{2}, \frac{1}{3}$

II. $6y^2 - 29y + 9 = 0$

$6y^2 - 27y - 2y + 9 = 0$

$3y(2y - 9) - 1(2y - 9) = 0$

$(3y - 1)(2y - 9) = 0$

$y = \frac{9}{2}, \frac{1}{3}$

Hence, $x \leq y$

Sol-39. 1

I. $20y^2 - 13y + 2 = 0$

$20y^2 - 8y - 5y + 2 = 0$

$4y(5y - 2) - 1(5y - 2) = 0$

$(4y - 1)(5y - 2) = 0$

$y = \frac{2}{5}, \frac{1}{4}$

II. $6x^2 - 25x + 14 = 0$

$6x^2 - 21x - 4x + 14 = 0$

$3x(2x - 7) - 2(2x - 7) = 0$

$(3x - 2)(2x - 7) = 0$

$x = \frac{7}{2}, \frac{2}{3}$

Hence, $x > y$

Assignment On Quadratic Equation

Sol-40. 2

I. $2x^2 - 9x + 9 = 0$

$2x^2 - 3x - 6x + 9 = 0$

$x(2x - 3) - 3(2x - 3) = 0$

$(x - 3)(2x - 3) = 0$

$x = 3, \frac{3}{2}$

II. $2y^2 - 7y + 3 = 0$

$2y^2 - y - 6y + 3 = 0$

$y(2y - 1) - 3(2y - 1) = 0$

$y = 3, \frac{1}{2}$

Hence, $x \geq y$

Sol-41. 1

I. $x^2 = 10$

$\therefore x = \sqrt{10} = \pm 3.16$

II. $y^2 - 9y + 20 = 0$

$y^2 - 5y - 4y + 20 = 0$

$y(y - 5) - 4(y - 5) = 0$

$(y - 4)(y - 5) = 0$

$\therefore y = 4, 5$

Hence $x < y$.

Assignment On Quadratic Equation

Sol-42. 3

I. $2x^2 - 15x + 27 = 0$

$2x^2 - 6x - 9x + 27 = 0$

$2x(x - 3) - 9(x - 3) = 0$

$(2x - 9)(x - 3) = 0$

$\therefore x = \frac{9}{2}, 3$

II. $2y^2 - 13y + 20 = 0$

$2y^2 - 8y - 5y + 20 = 0$

$2y(y - 4) - 5(y - 4) = 0$

$(2y - 5)(y - 4) = 0$

$\therefore y = \frac{5}{2}, 4$

Hence no relationship can be established.

Sol-43. 1

I. $9x^2 - 21x + 10 = 0$

$9x^2 - 6x - 15x + 10 = 0$

$3x(3x - 2) - 5(3x - 2) = 0$

$(3x - 5)(3x - 2) = 0$

$\therefore x = \frac{5}{3}, \frac{2}{3}$

II. $y^2 - 8y + 15 = 0$

$y^2 - 5y - 3y + 15 = 0$

$y(y - 5) - 3(y - 5) = 0$

$(y - 5)(y - 3) = 0$

$\therefore y = 3, 5$

Hence $x < y$

Sol-44. 3

I. $2x^2 - 13x + 15 = 0$

$$2x^2 - 10 - 3 + 15 = 0$$

$$2x(x - 5) - 3(x - 5) = 0$$

$$(2x - 3)(x - 5) = 0$$

$$\therefore x = \frac{3}{2}, 5$$

II. $2y^2 - 11y + 12 = 0$

$$2y^2 - 8y - 3y + 12 = 0$$

$$2y(y - 4) - 3(y - 4) = 0$$

$$(2y - 3)(y - 4) = 0$$

$$\therefore y = \frac{3}{2}, 4$$

Hence no relationship can be established.

Sol-45. 2

I. $2x^2 + 7x + 6 = 0$

$$2x^2 + 4x + 3x + 6 = 0$$

$$2x(x + 2) + 3(x + 2) = 0$$

$$(2x + 3)(x + 2) = 0$$

$$\therefore x = \frac{-3}{2}, -2$$

II. $2y^2 + 17y + 30 = 0$

$$2y^2 + 12y + 5y + 30 = 0$$

$$2y(y + 6) + 5(y + 6) = 0$$

$$(2y + 5)(y + 6) = 0$$

$$\therefore y = -6, \frac{-5}{2}$$

Hence $x > y$.

Sol-46.: 4

I. $8x^2 - 14x + 3 = 0$

$8x^2 - 2x - 12x + 3 = 0$

$2x(4x - 1) - 3(4x - 1) = 0$

$(2x - 3)(4x - 1) = 0$

$\therefore x = \frac{1}{4}, \frac{3}{2}$

II. $8y^2 - 22y + 5 = 0$

$8y^2 - 20y - 2y + 5 = 0$

$4y(2y - 5) - 1(2y - 5) = 0$

$(2y - 5)(4y - 1) = 0$

$\therefore y = \frac{1}{4}, \frac{5}{2}$

Hence, relationship cannot be established between x and y

Sol-47. 1

I. $2x^2 - 11x + 12 = 0$

$2x^2 - 8x - 3x + 12 = 0$

$2x(x - 4) - 3(x - 4) = 0$

$(2x - 3)(x - 4) = 0$

$\therefore x = \frac{3}{2}, 4$

II. $12y^2 + 29y + 15 = 0$

$12y^2 + 20y + 9y + 15 = 0$

$4y(3y + 5) + 3(4y + 5) = 0$

$(3y + 5)(4y + 3) = 0$

$\therefore y = -\frac{3}{4}, -\frac{5}{3}$

Hence, $x > y$

Sol-48. 5

I. $2y^2 - 3y - 14 = 0$

$2y^2 - 7y + 4y - 14 = 0$

$y(2y - 7) + 2(2y - 7) = 0$

$(y + 2)(2y - 7) = 0$

$\therefore y = -2, \frac{7}{2}$

II. $3x^2 - 7x + 4 = 0$

$3x^2 - 4x - 3x + 4 = 0$

$x(3x - 4) - 1(3x - 4) = 0$

$\therefore x = 1, \frac{4}{3}$

Hence, relationship cannot be established between x and y

Sol-49. 3

I. $6x^2 + 37x + 45 = 0$

$6x^2 + 10x + 27x + 45 = 0$

$2x(3x + 5) + 9(3x + 5) = 0$

$(2x + 9)(3x + 5) = 0$

$\therefore x = -\frac{9}{2}, -\frac{5}{3}$

II. $3y^2 - 11y + 6 = 0$

$3y^2 - 2y - 9y + 6 = 0$

$y(3y - 2) - 3(3y - 2) = 0$

$(y - 3)(3y - 2) = 0$

$\therefore y = 3, \frac{2}{3}$

Hence, $x < y$

Assignment On Quadratic Equation

Sol-50. 5

I. $7x^2 + 52x + 21 = 0$

$$7x^2 + 3x + 49x + 21 = 0$$

$$x(7x + 3) + 7(7x + 3) = 0$$

$$(x + 7)(7x + 3) = 0$$

$$\therefore x = -7, -\frac{3}{7}$$

II. $6y^2 + 7y - 24 = 0$

$$6y^2 + 16y - 9y - 24 = 0$$

$$2y(3y + 8) - 3(3y + 8) = 0$$

$$(2y - 3)(3y + 8) = 0$$

$$\therefore y = \frac{3}{2}, -\frac{8}{3}$$

Hence, relation cannot be established between x and y

Sol-51. 5

I. $35x^2 - 24x - 35 = 0$

$$35x^2 + 25x - 49x - 35 = 0$$

$$5x(7x + 5) - 7(7x + 5) = 0$$

$$(5x - 7)(7x + 5) = 0$$

$$\therefore x = \frac{7}{5}, -\frac{5}{7}$$

II. $72y^2 - 145y + 72 = 0$

$$72y^2 - 64y - 81y + 72 = 0$$

$$8y(9y - 8) - 9(9y - 8) = 0$$

$$(8y - 9)(9y - 8) = 0$$

$$\therefore y = \frac{9}{8}, \frac{8}{9}$$

Hence, relationship cannot be established between x and y .

Sol-52. 5

I. $77x^2 - 25x - 72 = 0$

$77x^2 - 88x + 63x - 72 = 0$

$11x(7x - 8) + 9(7x - 8) = 0$

$(11x + 9)(7x - 8) = 0$

$x = -\frac{9}{11}, \frac{8}{7}$

II. $42y^2 + 13y - 42 = 0$

$42y^2 + 49y - 36y - 42 = 0$

$7y(6y + 7) - 6(6y + 7) = 0$

$(7y - 6)(6y + 7) = 0$

$y = \frac{6}{7}, -\frac{7}{6}$

Hence, relationship cannot be established between x and y

Sol-53. 2

I. $35x^2 - 51x + 18 = 0$

$35x^2 - 21x - 30x + 18 = 0$

$7x(5x - 3) - 6(5x - 3) = 0$

$(7x - 6)(5x - 3) = 0$

$x = \frac{6}{7}, \frac{3}{5}$

II. $30y^2 + 17y - 21 = 0$

$30y^2 + 35y - 18y - 21 = 0$

$5y(6y + 7) - 3(6y + 7) = 0$

$(5y - 3)(6y + 7) = 0$

$y = \frac{3}{5}, -\frac{7}{6}$

Hence, $x \geq y$

Assignment On Quadratic Equation

Sol- 54. 3

$$I. 117x^2 + 250x + 117 = 0$$

$$117x^2 + 169x + 81x + 117 = 0$$

$$13x(9x + 13) + 9(9x + 13) = 0$$

$$(13x + 9)(9x + 13) = 0$$

$$x = -\frac{9}{13}, -\frac{13}{9}$$

$$II. 54y^2 - 123y + 65 = 0$$

$$54y^2 - 45y - 78y + 65 = 0$$

$$9y(6y - 5) - 13(6y - 5) = 0$$

$$(9y - 13)(6y - 5) = 0$$

$$y = \frac{13}{9}, \frac{5}{6}$$

Hence, $x < y$

Sol-55. 3

$$I. 56x^2 - 99x + 40 = 0$$

$$56x^2 - 35x - 64x + 40 = 0$$

$$7x(8x - 5) - 8(8x - 5) = 0$$

$$(7x - 8)(8x - 5) = 0$$

$$x = \frac{8}{7}, \frac{5}{8}$$

$$II. 8y^2 - 30y + 25 = 0$$

$$8y^2 - 10y - 20y + 25 = 0$$

$$2y(4y - 5) - 5(4y - 5) = 0$$

$$(2y - 5)(4y - 5) = 0$$

$$y = \frac{5}{2}, \frac{5}{4}$$

Hence $x < y$

Assignment On Quadratic Equation

Sol-56. 5

I. $4x^2 - 21x + 20 = 0$

$4x^2 - 16x - 5x + 20 = 0$

$4x(x - 4) - 5(x - 4) = 0$

$(4x - 5)(x - 4) = 0$

$\therefore x = 4, \frac{5}{4}$

II. $8y^2 - 22y + 15 = 0$

$8y^2 - 12y - 10y + 15 = 0$

$4y(2y - 3) - 5(2y - 3) = 0$

$(4y - 5)(2y - 3) = 0$

$\therefore y = \frac{5}{4}, \frac{3}{2}$

Hence, relationship cannot be established between x and y

Sol-57. 4

I. $y^2 - 7y - 18 = 0$

$y^2 + 2y - 9y - 18 = 0$

$y(y + 2) - 9(y + 2) = 0$

$(y + 2)(y - 9) = 0$

$\therefore y = -2, 9$

II. $x^2 + 10x + 16 = 0$

$x^2 + 8x + 2x + 16 = 0$

$x(x + 8) + 2(x + 8) = 0$

$x = -8, -2$

Hence, $x \leq y$

Sol-58. 3

I. $2y^2 - 11y + 15 = 0$

$2y^2 - 5y - 6y + 15 = 0$

$y(2y - 5) - 3(2y - 5) = 0$

$(y - 3)(2y - 5) = 0$

$y = 3, \frac{5}{2}$

II. $2x^2 + 3x - 14 = 0$

$2x^2 + 7x - 4x - 14 = 0$

$x(2x + 7) - 2(2x + 7) = 0$

$(x - 2)(2x + 7) = 0$

$x = 2, -\frac{7}{2}$

Hence, $x < y$

Sol-59. 3

I. $3x^2 - 5x + 2 = 0$

$3x^2 - 3x - 2x + 2 = 0$

$3x(x - 1) - 2(x - 1) = 0$

$(3x - 2)(x - 1) = 0$

$x = 1, \frac{2}{3}$

II. $6y^2 - 19y + 14 = 0$

$6y^2 - 12y - 7y + 14 = 0$

$6y(y - 2) - 7(y - 2) = 0$

$(6y - 7)(y - 2) = 0$

$y = 2, \frac{7}{6}$

Hence, $x < y$

Assignment On Quadratic Equation

Sol-60. 5

I. $6y^2 + 11y - 7 = 0$

$6y^2 + 14y - 3y - 7 = 0$

$2y(3y + 7) - 1(3y + 7) = 0$

$(2y - 1)(3y + 7) = 0$

$y = \frac{1}{2}, -\frac{7}{3}$

II. $21x^2 + 5x - 6 = 0$

$21x^2 + 14x - 9x - 6 = 0$

$7x(3x + 2) - 3(3x + 2) = 0$

$(7x - 3)(3x + 2) = 0$

$x = \frac{3}{7}, -\frac{2}{3}$

Hence, relationship cannot be established between x and y

Sol-61. 3

I. $17x^2 - 26x - 16 = 0$

$17x^2 - 34x + 8x - 16 = 0$

$17x(x - 2) + 8(x - 2) = 0$

$(17x + 8)(x - 2) = 0$

$x = 2, -\frac{8}{17}$

II. $17y^2 - 26y + 9 = 0$

$17y^2 - 17y - 9y + 9 = 0$

$17y(y - 1) - 9(y - 1) = 0$

$y = 1, \frac{9}{17}$

Hence, no relationship cannot be established between x and y

Sol-62 -3

I. $24x^2 - 58x + 23 = 0$

$24x^2 - 46x - 12x + 23 = 0$

$2x(12x-23) - 1(12x-23) = 0$

$(12x-23)(2x-1) = 0$

$x = \frac{1}{2}, \frac{23}{12}$

II. $20y^2 + 24y - 65 = 0$

$20y^2 + 50y - 26y - 65 = 0$

$10y(2y+5) - 13(2y+5) = 0$

$(2y+5)(10y-13) = 0$

$y = -\frac{5}{2}, \frac{13}{10}$

Hence, no relationship cannot be established between x and y .

Sol-63. 1

I. $x^{\frac{7}{3}} \div 27 = 512 \div x^{\frac{2}{3}}$

$\frac{x^{\frac{7}{3}}}{27} = \frac{512}{x^{\frac{2}{3}}}$

$x^{\frac{9}{3}} = 512 \times 27$

$x^3 = 13824$

$x = 24$

II. $y^{\frac{1}{5}} \times y^{\frac{1}{5}} \times 16 = 1296 \div y^{\frac{3}{5}}$

$y^{\frac{2}{5}} \times 16 = \frac{1296}{y^{\frac{3}{5}}}$

$y^{\frac{2}{5}} \times y^{\frac{3}{5}} = \frac{1296}{16}$

$y = 81$

Hence, $x < y$

Assignment On Quadratic Equation

Sol-64. 3

Let $x^3 = b$

$$3b^2 - 19b + 16 = 0$$

$$3b^2 - 16b - 3b + 16 = 0$$

$$(3b - 16)(b - 1) = 0$$

$$b = 1, \frac{16}{3}$$

$$x^3 = 1, \frac{16}{3}$$

$$x = 1, \sqrt[3]{\frac{16}{3}}$$

Let $y^2 = a$

$$9a^2 - 27a + 20 = 0$$

$$9a^2 - 15a - 12a + 20 = 0$$

$$(3a - 4)(3a - 5) = 0$$

$$a = \frac{4}{3}, \frac{5}{3}$$

$$y^2 = \frac{4}{3}, \frac{5}{3}$$

$$y = \frac{2}{\sqrt{3}}, \frac{\sqrt{5}}{\sqrt{3}}$$

Hence, the relationship cannot be established

Assignment On Quadratic Equation

Sol-65. 2

$$\sqrt{2x+9} + x = 13$$

$$\sqrt{2x+9} = 13 - x$$

Squaring both the sides,

$$2x + 9 = 169 + x^2 - 26x$$

$$x^2 - 28x + 160 = 0$$

$$(x - 8)(x - 20) = 0$$

$$x = 8, 20$$

$$\sqrt{4y^2 - 4y + 1} - 2y + 6 = 0$$

$$\sqrt{4y^2 - 4y + 1} = 2y - 6$$

Squaring both the sides

$$4y^2 - 4y + 1 = 4y^2 + 36 - 24y$$

$$20y = 35$$

$$y = \frac{7}{4}$$

Hence, $x > y$

Sol-66. 2

I. $x = \sqrt{16102 - 3333}$

$$x = \sqrt{12769} = 113$$

II. $y^2 = 12769$

$$y = \pm 113$$

Hence, $x \geq y$

B Sol-67. 5

I. $4x^2 + 3\sqrt{7}x - 7 = 0$

$$4x^2 + 4\sqrt{7}x - \sqrt{7}x - 7 = 0$$

$$4x(x + \sqrt{7}) - \sqrt{7}(x + \sqrt{7}) = 0$$

$$(x + \sqrt{7})(4x - \sqrt{7}) = 0$$

$$x = -\sqrt{7}, \frac{\sqrt{7}}{4}$$

Or

$$x = \frac{-7\sqrt{7}}{7}, \frac{\sqrt{7}}{4}$$

II. $7y^2 + 4\sqrt{7}y - 5 = 0$

$$7y^2 + 5\sqrt{7}y - \sqrt{7} - 5 = 0$$

$$\sqrt{7}y(\sqrt{7}y + 5) - 1(\sqrt{7}y + 5) = 0$$

$$(\sqrt{7}y + 5)(\sqrt{7}y - 1) = 0$$

$$y = \frac{-5}{\sqrt{7}}, \frac{1}{\sqrt{7}}$$

Or

$$y = \frac{-5\sqrt{7}}{7}, \frac{\sqrt{7}}{7}$$

Hence, No relation can be established between x and y

Sol-68. 3

By multiplying Equation I by 11 and Multiplying Equation II by 7 and solving the equation we get,

$$77x + 121y = 1551$$

$$-77x + 63y = 105$$

$$184y = 1656$$

$$y = 9$$

By putting y in the value in equation I

$$x = 6$$

Hence, $x < y$

Sol-69. 5

$$I. 6x^2 - 41x + 13 = 0$$

$$6x^2 - 39x - 2x + 13 = 0$$

$$3x(2x - 13) - 1(2x - 13) = 0$$

$$(2x - 13)(3x - 1) = 0$$

$$x = \frac{13}{2}, \frac{1}{3}$$

$$II. 2y^2 - 19y + 42 = 0$$

$$2y^2 - 12y - 7y + 42 = 0$$

$$2y(y - 6) - 7(y - 6) = 0$$

$$(y - 6)(2y - 7) = 0$$

$$y = 6, \frac{7}{2}$$

Hence, no relation between x and y can be established.

Sol-70. 4

$$I. x = \sqrt{20 + \sqrt{20 + \sqrt{20 + \sqrt{20 \dots \dots \dots \infty}}}}$$

$$x = \sqrt{20 + x}$$

By squaring both sides,

$$x^2 = 20 + x$$

$$x^2 - x - 20 = 0$$

$$x^2 - 5x + 4x - 20 = 0$$

$$x(x - 5) + 4(x - 5) = 0$$

$$x = 5, -4$$

$$II. y = \sqrt{5\sqrt{5\sqrt{5\sqrt{5 \dots \dots \dots \infty}}}}$$

$$y = \sqrt{5y}$$

By squaring both sides,

$$y^2 = 5y$$

$$y = 5$$

Hence, $x \leq y$

Assignment On Quadratic Equation

Sol-71. 4

I. $\sqrt{289} x = -\sqrt{25}$

Squaring both sides,

$$289 x = 25 \Rightarrow \frac{25}{289} = 0.08$$

II. $\sqrt{676} y = -10$

Squaring both sides,

$$676 y = 100 \Rightarrow \frac{100}{676} = 0.14$$

Clearly, $x < y$

Sol-72. 1

I. $8x^2 - 78x + 169 = 0$

$$(4x - 13)(2x - 13) = 0$$

$$\Rightarrow x = \frac{13}{4}, \frac{13}{2}$$

II. $20y^2 - 117y + 169 = 0$

$$(4y - 13)(5y - 13) = 0$$

$$\Rightarrow y = \frac{13}{4}, \frac{13}{5}$$

Hence, $x \geq y$

Sol-73. 3

$$I. \frac{15+9}{\sqrt{x}} = 11\sqrt{x}$$

$$\frac{24}{\sqrt{x}} = 11\sqrt{x}$$

$$\Rightarrow 24 = 11x$$

$$\Rightarrow x = \frac{24}{11}$$

$$II. \frac{3\sqrt{y} + 5\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$$

$$\frac{8\sqrt{y}}{12} = \frac{1}{\sqrt{y}} \Rightarrow 8y = 12$$

$$\Rightarrow y = \frac{12}{8}$$

Hence, $x > y$

Sol-74. 5

$$I. \frac{8}{\sqrt{x}} + \frac{6}{\sqrt{x}} = \sqrt{x} \Rightarrow \frac{8+6}{\sqrt{x}} = \sqrt{x}$$

$$\Rightarrow \frac{14}{\sqrt{x}} = \sqrt{x} \Rightarrow 14 = x$$

$$\Rightarrow x = 14$$

$$II. y^3 - \frac{(14)^{\frac{7}{2}}}{\sqrt{y}} = 0$$

$$\Rightarrow y^3 = \frac{(14)^{\frac{7}{2}}}{\sqrt{y}} \Rightarrow (y)^{\frac{7}{2}} = (14)^{\frac{7}{2}}$$

$$\Rightarrow y = 14$$

Hence, $x = y$

Assignment On Quadratic Equation

Sol-75. 5

I. $x^2 = 441 \Rightarrow x = \pm 21$

II. $y^2 = 371 - 47 = \pm 18$

Hence, relationship cannot be established between x and y .

Sol-76. 1

I. $90x^2 - 113x + 35 = 0$

$$90x^2 - 50x - 63x + 35 = 0$$

$$10x(9x - 5) - 7(9x - 5) = 0$$

$$(10x - 7)(9x - 5) = 0$$

$$x = \frac{7}{10}, \frac{5}{9}$$

II. $88y^2 - 149y + 63 = 0$

$$88y^2 - 72y - 77y + 63 = 0$$

$$8y(11y - 9) - 7(11y - 9) = 0$$

$$(8y - 7)(11y - 9) = 0$$

$$y = \frac{7}{8}, \frac{9}{11}$$

Hence $x < y$.

Sol-77. 3

I. $35x^2 + 83x + 36 = 0$

$$35x^2 + 63x + 20x + 36 = 0$$

$$7x(5x + 9) + 4(5x + 9) = 0$$

$$(7x + 4)(5x + 9) = 0$$

$$x = -\frac{4}{7}, -\frac{9}{5}$$

II. $42y^2 + 53y + 15 = 0$

$$42y^2 + 18y + 35y + 15 = 0$$

$$6y(7y + 3) + 5(7y + 3)$$

$$(6y + 5)(7y + 3) = 0$$

$$y = -\frac{5}{6}, -\frac{3}{7}$$

Hence no relationship can be established.

Sol-78. 3

I. $96x^2 + 52x - 63 = 0$

$$96x^2 + 108x - 56x - 63 = 0$$

$$12x(8x + 9) - 7(8x + 9) = 0$$

$$(12x - 7)(8x + 9) = 0$$

$$x = \frac{7}{12}, -\frac{9}{8}$$

II. $77y^2 + 155y + 72 = 0$

$$77y^2 + 56y + 99y + 72 = 0$$

$$7y(11y + 8) + 9(11y + 8)$$

$$(7y + 9)(11y + 8) = 0$$

$$y = -\frac{9}{7}, -\frac{8}{11}$$

Hence no relationship can be established.

Assignment On Quadratic Equation

Sol-79. 2

I. $20x^2 - 93x + 108 = 0$

$$20x^2 - 48x - 45x + 108 = 0$$

$$4x(5x - 12) - 9(5x - 12) = 0$$

$$(4x - 9)(5x - 12) = 0$$

$$x = \frac{9}{4}, \frac{12}{5}$$

II. $72y^2 - 47y - 144 = 0$

$$72y^2 + 81y - 128y - 144 = 0$$

$$9y(8y + 9) - 16(8y + 9) = 0$$

$$(9y - 16)(8y + 9) = 0$$

$$y = \frac{16}{9}, -\frac{9}{8}$$

Hence $x > y$.

Sol-80. 1

Solution:

I. $9x^2 + 6\sqrt{7}x + 7 = 0$

$$(3x + \sqrt{7})^2 = 0$$

$$3x = -\sqrt{7}$$

$$x = -\frac{\sqrt{7}}{3}$$

II. $4y^2 + 4\sqrt{3}y + 3 = 0$

$$(2y + \sqrt{3})^2 = 0$$

$$2y = -\sqrt{3}$$

$$y = -\frac{\sqrt{3}}{2}$$

Hence $x < y$.

Sol-81. 5

I. $143y^2 - 161y + 36 = 0$

$143y^2 - 117y - 44y + 36 = 0$

$13y(11y - 9) - 4(11y - 9) = 0$

$(13y - 4)(11y - 9) = 0$

$\therefore y = \frac{4}{13}, \frac{9}{11}$

II. $36x^2 - 161x + 143 = 0$

$36x^2 - 117x - 44x + 143 = 0$

$9x(4x - 13) - 11(4x - 13) = 0$

$(9x - 11)(4x - 13) = 0$

$\therefore x = \frac{11}{9}, \frac{13}{4}$

Hence, $x > y$

Sol-82. 5

I. $84y^2 - 187y + 104 = 0$

$84y^2 - 91y - 96y + 104 = 0$

$7y(12y - 13) - 8(12y - 13) = 0$

$(7y - 8)(12y - 13) = 0$

$\therefore y = \frac{8}{7}, \frac{13}{12}$

II. $132x^2 + 41x - 40 = 0$

$132x^2 + 96x - 55x - 40 = 0$

$12x(11x + 8) - 5x(11x + 8) = 0$

$(12x - 5)(11x + 8) = 0$

$\therefore x = \frac{5}{12}, \frac{-8}{11}$

Hence, $x < y$

Sol-83. 5

I. $63x^2 + 146x + 80 = 0$

$63x^2 + 90x + 56x + 80 = 0$

$9x(7x + 10) + 8(7x + 10) = 0$

$(9x + 8)(7x + 10) = 0$

$\therefore x = \frac{-8}{9}, \frac{-10}{7}$

II. $42y^2 + 109y + 70 = 0$

$42y^2 + 60y + 49y + 70 = 0$

$6y(7y + 10) + 7(7y + 10) = 0$

$(6y + 7)(7y + 10) = 0$

$\therefore y = \frac{-7}{6}, \frac{-10}{7}$

Hence, $x \geq y$

Sol-84. 5

I. $70x^2 - 227x + 117 = 0$

$70x^2 - 45x - 182x + 117 = 0$

$5x(19x - 9) - 13(14x - 9) = 0$

$(5x - 13)(14x - 9) = 0$

$\therefore x = \frac{13}{5}, \frac{9}{14}$

II. $72y^2 + 23y - 35 = 0$

$72y^2 - 40y + 63y - 35 = 0$

$8y(9y - 5) + 7(9y - 5) = 0$

$(8y + 7)(9y - 5) = 0$

$\therefore y = \frac{-7}{8}, \frac{5}{9}$

Hence, $x > y$

Sol-85. 5

I. $40x^2 + 81x + 35 = 0$

$40x^2 + 56x + 25x + 35 = 0$

$8x(5x + 7) + 5(5x + 7) = 0$

$(8x + 5)(5x + 7) = 0$

$\therefore x = \frac{-5}{8}, \frac{-7}{5}$

II. $63y^2 + 103y + 42 = 0$

$63y^2 + 49y + 54y + 42 = 0$

$7y(9y + 7) + 6(9y + 7) = 0$

$(7y + 6)(9y + 7) = 0$

$\therefore y = \frac{-6}{7}, \frac{-7}{9}$

Hence, relationship cannot be established between x and y

Sol-86. 5

I. $35x^2 - 46x - 16 = 0$

$35x^2 - 56x + 10x - 16 = 0$

$7x(5x - 8) + 2(5x - 8) = 0$

$(7x + 2)(5x - 8) = 0$

$\therefore x = -\frac{2}{7}, \frac{8}{5}$

II. $35y^2 - 116y + 96 = 0$

$35y^2 - 56y - 60y + 96 = 0$

$7y(5y - 8) - 12(5y - 8) = 0$

$(7y - 12)(5y - 8) = 0$

$\therefore y = \frac{12}{7}, \frac{8}{5}$

Hence, $x \leq y$

Sol-87. 3

I. $195x^2 - 46x - 21 = 0$

$195x^2 + 45x - 91x - 21 = 0$

$15x(13x + 3) - 7(13x + 3) = 0$

$(13x + 3)(15x - 7) = 0$

$\therefore x = \frac{7}{15}, -\frac{3}{13}$

II. $209y^2 + 7y - 12 = 0$

$209y^2 + 57y - 44y - 12 = 0$

$19y(11y + 3) - 4(11y + 3) = 0$

$(19y - 4)(11y + 3) = 0$

$\therefore y = \frac{4}{19}, -\frac{3}{11}$

Hence, relationship cannot be established between x and y

Sol-88. 1

I. $165x^2 + 97x + 10 = 0$

$165x^2 + 75x + 22x + 10 = 0$

$11x(15x + 2) + 5(15x + 2) = 0$

$(11x + 5)(15x + 2) = 0$

$\therefore x = -\frac{5}{11}, -\frac{2}{15}$

II. $117y^2 - 163y + 56 = 0$

$117y^2 - 72y - 91y + 56 = 0$

$9y(13y - 8) - 7(13y - 8) = 0$

$(9y - 7)(13y - 8) = 0$

$\therefore y = \frac{7}{9}, \frac{8}{13}$

Hence, $\therefore x < y$

Sol-89. 3

I. $144x^2 - 163x - 65 = 0$

$144x^2 - 208x + 45x - 65 = 0$

$16x (9x - 13) + 5(9x - 13) = 0$

$(16x + 5) (9x - 13) = 0$

$\therefore x = -\frac{5}{16}, \frac{13}{9}$

II. $91y^2 - 128y - 48 = 0$

$91y^2 + 28y - 156y - 48 = 0$

$7y (13y + 4) - 12 (13y + 4) = 0$

$(13y + 4) (7y - 12) = 0$

$\therefore y = \frac{12}{7}, -\frac{4}{13}$

Hence, relationship cannot be established between x and y

Sol-90. 2

I. $84x^2 - 167x - 55 = 0$

$84x^2 - 35x - 132x - 55 = 0$

$7x (12x - 5) - 11 (12x - 5) = 0$

$(12x - 5) (7x - 11) = 0$

$\therefore x = \frac{5}{12}, \frac{11}{7}$

II. $247y^2 + 210y + 27 = 0$

$247y^2 + 171y + 39y + 27 = 0$

$19y (13y + 9) + 3(13y + 9) = 0$

$(19y + 3) (13y + 9) = 0$

$\therefore y = -\frac{3}{19}, -\frac{9}{13}$

Hence, $x > y$

Sol-91. 3

I. $27x^2 + 120x + 77 = 0$

$27x^2 + 21x + 99x + 77 = 0$

$3x(9x + 7) + 11(9x + 7) = 0$

$(9x + 7)(3x + 11) = 0$

$\therefore x = -\frac{7}{9}, -\frac{11}{3}$

II. $56y^2 + 117y + 36 = 0$

$56y^2 + 96y + 21y + 36 = 0$

$8y(7y + 12) + 3(7y + 12) = 0$

$(8y + 3)(7y + 12) = 0$

$\therefore y = -\frac{3}{8}, -\frac{12}{7}$

Hence, no relationship cannot be established between x and y .

Sol-92. 1

I. $104x^2 + 9x - 35 = 0$

$104x^2 + 65x - 56x - 35 = 0$

$13x(8x + 5) - 7(8x + 5) = 0$

$(13x - 7)(8x + 5) = 0$

$\therefore x = \frac{7}{13}, -\frac{5}{8}$

II. $72y^2 - 85y + 25 = 0$

$72y^2 - 40y - 45y + 25 = 0$

$8y(9y - 5) - 5(9y - 5) = 0$

$(8y - 5)(9y - 5) = 0$

$\therefore y = \frac{5}{8}, \frac{5}{9}$

Hence, $x < y$

Sol-93. 3

I. $99x^2 + 161x + 26 = 0$

$99x^2 + 143x + 18x + 26 = 0$

$11x(9x + 13) + 2(9x + 13) = 0$

$(11x + 2)(9x + 13) = 0$

$\therefore x = -\frac{2}{11}, -\frac{13}{9}$

II. $26y^2 + 161y + 99 = 0$

$26y^2 + 18y + 143y + 99 = 0$

$2y(13y + 9) + 11(13y + 9) = 0$

$\therefore y = -\frac{11}{2}, -\frac{9}{13}$

Hence, relationship cannot be established between x and y .

Sol-94. 4

I. $70x^2 - 143x + 72 = 0$

$70x^2 - 63x - 80x + 72 = 0$

$7x(10x - 9) - 8(10x - 9) = 0$

$(7x - 8)(10x - 9) = 0$

$\therefore x = \frac{9}{10}, \frac{8}{7}$

II. $80y^2 - 142y + 63 = 0$

$80y^2 - 70y - 72y + 63 = 0$

$10y(8y - 7) - 9(8y - 7) = 0$

$(10y - 9)(8y - 7) = 0$

$\therefore y = \frac{9}{10}, \frac{7}{8}$

Hence, $x \geq y$.

Sol-95. 1

I. $99x^2 + 31x - 110 = 0$

I. $99x^2 + 121x - 90x - 110 = 0$

$11x(9x + 11) - 10(9x + 11) = 0$

$(11x - 10)(9x + 11) = 0$

$\therefore x = \frac{10}{11}, -\frac{11}{9}$

II. $6y^2 - 31y + 35 = 0$

$6y^2 - 10y - 21y + 35 = 0$

$2y(3y - 5) - 7(3y - 5) = 0$

$(2y - 7)(3y - 5) = 0$

$\therefore y = \frac{7}{2}, \frac{5}{3}$

Hence, $x < y$

Sol-96. 5

I. $2x^2 - 12x + 16 = 0$

$2x^2 - 8x - 4x + 16 = 0$

$2x(x - 4) - 4(x - 4)$

$(2x - 4)(x - 4)$

$x = 4, 2$

II. $4y^2 - 8y - 12 = 0$

$4y^2 - 12y + 4y - 12 = 0$

$2y(y - 3) - 4(y - 3)$

$(2y - 4)(y - 3)$

$x = 2, 3$

Hence relationship between x and y cannot be established.

Sol-97. 4

$$I. 5x^2 - 24x + 28 = 0$$

$$5x^2 - 14x - 10x + 28 = 0$$

$$x(5x - 14) - 2(5x - 14)$$

$$(x - 2)(5x - 14)$$

$$x = \frac{14}{5}, 2$$

$$II. 2y^2 - y - 21 = 0$$

$$2y^2 - 7y - 6y - 21 = 0$$

$$y(2y - 7) - 3(2y - 7)$$

$$(y - 3)(2y - 7)$$

$$y = 3, \frac{7}{2}$$

Hence, $x < y$

Sol-98. 4

$$I. 5x^2 - 14x + 8 = 0$$

$$5x^2 - 10x - 4x + 8 = 0$$

$$5x(x - 2) - 4(5x - 2)$$

$$(x - 2)(5x - 4)$$

$$x = 2, \frac{4}{5}$$

$$II. 2y^2 - y - 21 = 0$$

$$2y^2 - 7y - 6y - 21 = 0$$

$$y(2y - 7) - 3(2y - 7)$$

$$(y - 3)(2y - 7)$$

$$y = 3, \frac{7}{2}$$

Hence, $x < y$

Sol-99. 3

I. $3x^2 - 22x + 40 = 0$

$$3x^2 - 12x - 10x + 40 = 0$$

$$3x(x - 4) - 10(x - 4)$$

$$(3x - 10)(x - 4)$$

$$x = \frac{10}{3}, 4$$

II. $4y^2 + 22y + 24 = 0$

$$4y^2 + 16y + 6y + 24 = 0$$

$$4y(y + 4) + 6(y + 4)$$

$$(4y + 6)(y + 4)$$

$$x = -4, -\frac{3}{2}$$

Hence, $x > y$

Sol-100. 3

I. $2x^2 + 11x + 15 = 0$

$$2x^2 + 6x + 5x + 15 = 0$$

$$2x(x + 3) + 5(x + 3)$$

$$(2x + 5)(x + 3)$$

$$x = \frac{5}{2}, -3$$

II. $2y^2 - 19y + 44 = 0$

$$2y^2 - 11y - 8y - 44 = 0$$

$$y(2y - 11) - 4(2y - 11)$$

$$(y - 4)(2y - 11)$$

$$x = 4, \frac{11}{2}$$

Hence, $x > y$